

2026-04-27-jackett-management-ui-and-system-db

Jackett Management UI & System Database — Implementation Plan

Revision: 1 **Last modified:** 2026-06-06T00:00:00Z

For agentic workers: REQUIRED SUB-SKILL: Use superpowers:subagent-driven-development (recommended) or superpowers:executing-plans to implement this plan task-by-task. Steps use checkbox (- []) syntax for tracking.

Goal: Ship a Go-canonical boba-jackett service on port 7189 with a SQLite system database (encrypted credentials), a full Angular Jackett management UI in the Boba dashboard, and reach the 13-item Definition-of-Done from the design spec with zero loose ends.

Architecture: New Go service in qBitTorrent-go/cmd/boba-jackett runs alongside the existing Python merge service. SQLite at config/boba.db is canonical for credentials/indexer state/catalog cache; .env is mirrored on every change; Jackett is configured live via its REST API. AES-256-GCM encrypts credential values at rest with BOBA_MASTER_KEY auto-generated into .env on first boot. Two new Angular routes (/jackett/credentials, /jackett/indexers) call the new Go service.

Tech Stack: Go 1.26 (existing), github.com/mattn/go-sqlite3, Go stdlib crypto/aes + crypto/cipher, net/http + gorilla/mux (matching existing Go service patterns), Angular 21 + Material (existing dashboard), go-swagger/swagger for OpenAPI generation, bandit/gosec for security static-analysis (already in CI).

Spec: docs/superpowers/specs/2026-04-27-jackett-management-ui-and-system-db-design.md

Predecessor: docs/superpowers/plans/2026-04-26-jackett-autoconfig-clean-rebuild.md (Layers 1-7 of Python autoconfig — shipped, will be reconciled in Task 43)

Conventions Used Throughout

- **Resource limits on every test invocation** (CONST-09): prefix with `GOMAXPROCS=2 nice -n 19 ionice -c 3`.
 - **Conventional Commits** with the body line
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<noreply@anthropic.com>.
 - **Local commits only** during implementation. Final push to all remotes happens in Task 53 (one batch).
 - **Real services only** for non-unit tests (CONST-11). Use the live Jackett container; do NOT mock it past Layer 1.
 - **Reproduction-before-fix** (CONST-32): if a bug surfaces, write a challenge that reproduces it FIRST, confirm fail, then fix, then re-run.
 - **Concurrent-safe containers** (CONST-13): any new shared map/slice in Go uses `digital.vasic.concurrency/pkg/safe` or equivalent — never bare `sync.Mutex + map`.
 - **No host power management** (CONST-33): the project's two challenge scripts (`no_suspend_calls_challenge.sh` + `host_no_auto_suspend_challenge.sh`) must pass after every commit batch. Already verified clean on 2026-04-27.
-

File Structure (locked-in decomposition)

New files to create

```
qBittorrent-go/
├─ cmd/boba-jackett/
│   └─ main.go                                # Service
entrypoint, wires bootstrap + HTTP server
├─ Dockerfile.jackett                        # New container
image for boba-jackett
├─ internal/
│   └─ db/
│       └─ conn.go                            # SQLite
connection (WAL mode, foreign_keys ON)
│   └─ crypto.go                             # AES-256-GCM
helpers
```

			crypto_test.go	
			migrate.go	# Embedded
migrations stepper				
			migrate_test.go	
			migrations/	
			└ 001_init.sql	# Full schema from
spec \$5				
			└ repos/	
			└ credentials.go	
			└ credentials_test.go	
			└ indexers.go	
			└ indexers_test.go	
			└ catalog.go	
			└ catalog_test.go	
			└ runs.go	
			└ runs_test.go	
			└ overrides.go	
			└ overrides_test.go	
			└ envfile/	
			└ parse.go	# .env reader
(handles comments, quotes, blanks)				
			└ parse_test.go	
			└ write.go	# Atomic write:
tmp + fsync + rename + dir-fsync				
			└ write_test.go	
			└ bootstrap/	
			└ bootstrap.go	# Master key
autogen + .env update + first-run import				
			└ bootstrap_test.go	
			└ import.go	# .env triple → DB
row				
			└ jackett/	
			└ client.go	# HTTP client
(session warmup, catalog, indexer config)				
			└ client_test.go	
			└ matcher.go	# Fuzzy + override
matcher (port from Python)				
			└ matcher_test.go	
			└ autoconfig.go	# Orchestrator
(DB-aware port from Python)				
			└ autoconfig_test.go	
			└ jackettapi/	
			└ auth_middleware.go	# admin/admin
session check (shared with Python)				
			└ credentials.go	# /api/v1/jackett/
credentials handlers				
			└ credentials_test.go	
			└ indexers.go	# /api/v1/jackett/
indexers handlers				
			└ indexers_test.go	
			└ catalog.go	# /api/v1/jackett/
catalog handlers				

```

|   |   |─ catalog_test.go
|   |   |─ runs.go                                # /api/v1/jackett/
autoconfig/runs handlers
|   |   |─ runs_test.go
|   |   |─ overrides.go                            # /api/v1/jackett/
overrides handlers
|   |   |─ overrides_test.go
|   |   |─ health.go                               # /healthz handler
|   |   |─ router.go                               # Mux setup with
redactor middleware
|   |   |─ openapi.go                              # Generated
OpenAPI 3.1 spec
|   |   |─ logging/
|   |   |─ redactor.go                             # io.Writer
wrapper that redacts known secrets
|   |   |─ redactor_test.go
└─ tests/
    |─ integration/
    |   └─ jackett_db_test.go                        # Real SQLite +
real .env, no Jackett
    |─ e2e/
    |   └─ jackett_management_test.go                # Real stack
    |─ security/
    |   └─ credential_leak_test.go                  # The leak-grep
test
    └─ contract/
        └─ openapi_test.go                          # Validate live
responses vs. OpenAPI

frontend/src/app/
└─ jackett/
    |─ jackett.module.ts
    |─ jackett-routing.module.ts
    |─ credentials/
    |   |─ credentials.component.ts
    |   |─ credentials.component.html
    |   |─ credentials.component.scss
    |   |─ credential-edit-dialog.component.ts
    |   └─ credentials.service.ts
    └─ indexers/
        |─ indexers.component.ts                    # Tab container
        |─ indexers.component.html
        |─ configured-tab.component.ts
        |─ catalog-tab.component.ts
        |─ history-tab.component.ts
        |─ indexer-add-dialog.component.ts
        |─ iptorrents-cookie-flow.component.ts
        └─ indexers.service.ts

challenges/scripts/
└─ cred_roundtrip_challenge.sh
└─ env_db_drift_challenge.sh

```

- |─ iptorrents_cookie_flow_challenge.sh
- |─ master_key_autogen_challenge.sh
- |─ credential_leak_grep_challenge.sh
- |─ nnmclub_native_plugin_clarification_challenge.sh
- |─ boba_db_file_perms_challenge.sh

docs/
 |─ BOBA_DATABASE.md # NEW: schema,
 master key, key rotation, recovery
 |─ issues/fixed/BUGFIXES.md # If not present,
 create + populate per Task 50

Files to modify

qBitTorrent-go/go.mod	# Add go-sqlite3
dep	
qBitTorrent-go/scripts/build.sh	# Add boba-jackett
to build targets	
docker-compose.yml	# Add boba-jackett
service block	
start.sh	# Add
ensure_boba_master_key + boba-jackett wait	
download-proxy/src/api/jackett.py	# REMOVE (replaced
by Go service)	
download-proxy/src/api/__init__.py	# REMOVE jackett
router include	
frontend/src/app/app.routes.ts	# Register /
jackett route	
frontend/src/app/app.html	# Add Jackett nav
entry	
docs/JACKETT_INTEGRATION.md	# Rewrite §"Auto-
Configuration" as DB-backed	
CLAUDE.md	# Update
Architecture + Port Map	
AGENTS.md	# Same updates as
CLAUDE.md	
docs/superpowers/plans/2026-04-26-jackett-autoconfig-clean-	
rebuild.md # Reconcile checkboxes	

Phase 1 — Foundations (Tasks 1-9)

Goal: Working Go service skeleton with SQLite + crypto + envfile primitives, all unit-tested, no HTTP server yet.

Task 1: Bootstrap project skeleton

Files: - Create: qBitTorrent-go/cmd/boba-jackett/main.go - Modify: qBitTorrent-go/go.mod - Modify: qBitTorrent-go/scripts/build.sh



Step 1: Create cmd/boba-jackett/main.go minimal stub

```
package main

import (
    "fmt"
    "os"
)

func main() {
    fmt.Fprintln(os.Stderr, "boba-jackett: not yet implemented")
    os.Exit(1)
}
```



Step 2: Add SQLite driver dependency

Run: cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 go get github.com/mattn/go-sqlite3@latest

Expected: go.mod and go.sum updated.



Step 3: Update build script

Edit qBitTorrent-go/scripts/build.sh, append after the webui-bridge build line:

```
echo "Building boba-jackett..."
cd "$ROOT_DIR" && go build -o "$BIN_DIR/boba-jackett" ./cmd/boba-jackett
```



Step 4: Verify build

Run: cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 bash scripts/build.sh

Expected: three binaries in bin/ (qbittorrent-proxy, webui-bridge, boba-jackett).



Step 5: Commit

```
git add qBitTorrent-go/cmd/boba-jackett/main.go qBitTorrent-go/
      go.mod qBitTorrent-go/go.sum qBitTorrent-go/scripts/
      build.sh
git commit -m "$(cat <<'EOF'
feat(boba-jackett): project skeleton + SQLite dep + build wiring

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EOF
)"
```

Task 2: AES-256-GCM crypto helpers

Files: - Create: qBitTorrent-go/internal/db/crypto.go - Test:
qBitTorrent-go/internal/db/crypto_test.go



Step 1: Write the failing test

Create internal/db/crypto_test.go:

```
package db

import (
    "bytes"
    "crypto/rand"
    "strings"
    "testing"
)

func key32() []byte {
    k := make([]byte, 32)
    _, _ = rand.Read(k)
    return k
}

func TestEncryptDecryptRoundTrip(t *testing.T) {
    k := key32()
    plain := "rutracker_password_xyz!@#"
    blob, err := Encrypt(k, plain)
    if err != nil {
        t.Fatalf("Encrypt: %v", err)
    }
    got, err := Decrypt(k, blob)
    if err != nil {
        t.Fatalf("Decrypt: %v", err)
    }
    if got != plain {
```

```

        t.Fatalf("round-trip mismatch: got %q want %q", got,
plain)
    }
}

func TestEncryptRejectsEmpty(t *testing.T) {
    if _, err := Encrypt(key32(), ""); err == nil {
        t.Fatalf("expected error encrypting empty plaintext")
    }
}

func TestEncryptRejectsBadKey(t *testing.T) {
    if _, err := Encrypt(make([]byte, 16), "x"); err == nil {
        t.Fatalf("expected error with 16-byte key")
    }
}

func TestDecryptDetectsTamper(t *testing.T) {
    k := key32()
    blob, _ := Encrypt(k, "secret")
    blob[len(blob)-1] ^= 0x01 // flip last bit
    if _, err := Decrypt(k, blob); err == nil {
        t.Fatalf("expected GCM auth failure on tampered
ciphertext")
    }
}

func TestNonceUniqueness(t *testing.T) {
    k := key32()
    seen := make(map[string]struct{}, 100000)
    for i := 0; i < 100000; i++ {
        blob, _ := Encrypt(k, "x")
        nonce := string(blob[:12])
        if _, dup := seen[nonce]; dup {
            t.Fatalf("nonce collision at iter %d", i)
        }
        seen[nonce] = struct{}{}
    }
}

func TestBlobShape(t *testing.T) {
    blob, _ := Encrypt(key32(), "x")
    if len(blob) < 12+16
        { // nonce(12) + GCM tag(16) + at least 1 byte
            t.Fatalf("blob too short: %d", len(blob))
        }
    if !bytes.Equal(blob[:12], blob[:12]) ||
        strings.Contains(string(blob), "x") {
        // just sanity – the test above already proves round-trip

```



```
}  
}
```



Step 2: Run test to verify FAIL

Run: `cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 go test -short -run Crypto ./internal/db/ -v` Expected: FAIL — Encrypt/Decrypt undefined.



Step 3: Implement internal/db/crypto.go

```
package db
```

```
import (  
    "crypto/aes"  
    "crypto/cipher"  
    "crypto/rand"  
    "errors"  
    "fmt"  
    "io"  
)
```

```
const nonceSize = 12
```

```
var (  
    ErrEmptyPlaintext = errors.New("crypto: empty plaintext")  
    ErrShortBlob      = errors.New("crypto: blob shorter than  
        nonce")  
)
```

```
func Encrypt(key []byte, plaintext string) ([]byte, error) {  
    if len(plaintext) == 0 {  
        return nil, ErrEmptyPlaintext  
    }  
    block, err := aes.NewCipher(key)  
    if err != nil {  
        return nil, fmt.Errorf("aes.NewCipher: %w", err)  
    }  
    gcm, err := cipher.NewGCM(block)  
    if err != nil {  
        return nil, fmt.Errorf("cipher.NewGCM: %w", err)  
    }  
    nonce := make([]byte, nonceSize)  
    if _, err := io.ReadFull(rand.Reader, nonce); err != nil {  
        return nil, fmt.Errorf("rand.Read: %w", err)  
    }  
    ct := gcm.Seal(nil, nonce, []byte(plaintext), nil)  
    out := make([]byte, 0, nonceSize+len(ct))  
    out = append(out, nonce...)
```

```

    out = append(out, ct...)
    return out, nil
}

func Decrypt(key []byte, blob []byte) (string, error) {
    if len(blob) < nonceSize {
        return "", ErrShortBlob
    }
    block, err := aes.NewCipher(key)
    if err != nil {
        return "", fmt.Errorf("aes.NewCipher: %w", err)
    }
    gcm, err := cipher.NewGCM(block)
    if err != nil {
        return "", fmt.Errorf("cipher.NewGCM: %w", err)
    }
    nonce, ct := blob[:nonceSize], blob[nonceSize:]
    pt, err := gcm.Open(nil, nonce, ct, nil)
    if err != nil {
        return "", fmt.Errorf("gcm.Open: %w", err)
    }
    return string(pt), nil
}

```



Step 4: Run tests to verify PASS

Run: `cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 go test -short -run Crypto ./internal/db/ -v` Expected: 5 PASS.



Step 5: Commit

```

git add qBitTorrent-go/internal/db/crypto.go qBitTorrent-go/
      internal/db/crypto_test.go
git commit -m "$(cat <<'EOF'
feat(boba-jackett): AES-256-GCM crypto helpers + tests

```

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```

EOF
)"

```

Task 3: SQLite schema migration 001_init.sql + stepper

Files: - Create: qBitTorrent-go/internal/db/migrations/001_init.sql - Create: qBitTorrent-go/internal/db/migrate.go - Create: qBitTorrent-go/internal/db/conn.go - Test: qBitTorrent-go/internal/db/migrate_test.go



Step 1: Create the migration SQL

Create internal/db/migrations/001_init.sql with the exact contents from spec §5 (reproduced verbatim — paste from the spec, do not re-derive).



Step 2: Write the failing migrate test

Create internal/db/migrate_test.go:

```
package db

import (
    "path/filepath"
    "testing"
)

func TestMigrateAppliesSchema(t *testing.T) {
    dir := t.TempDir()
    path := filepath.Join(dir, "test.db")
    conn, err := Open(path)
    if err != nil {
        t.Fatalf("Open: %v", err)
    }
    defer conn.Close()
    if err := Migrate(conn); err != nil {
        t.Fatalf("Migrate: %v", err)
    }
    wantTables := []string{"credentials", "indexers",
        "catalog_cache", "autoconfig_runs",
        "indexer_map_overrides", "schema_migrations"}
    for _, tbl := range wantTables {
        var name string
        err :=
            conn.QueryRow("SELECT name FROM sqlite_master WHERE
                type='table' AND name=?", tbl).Scan(&name)
        if err != nil {
            t.Errorf("table %s missing: %v", tbl, err)
        }
    }
}
```

```

func TestMigrateIdempotent(t *testing.T) {
    dir := t.TempDir()
    path := filepath.Join(dir, "test.db")
    conn, _ := Open(path)
    defer conn.Close()
    if err := Migrate(conn); err != nil {
        t.Fatalf("first migrate: %v", err)
    }
    if err := Migrate(conn); err != nil {
        t.Fatalf("second migrate (should be no-op): %v", err)
    }
    var version int
    if err := conn.QueryRow("SELECT MAX(version) FROM
        schema_migrations").Scan(&version); err != nil {
        t.Fatalf("query version: %v", err)
    }
    if version != 1 {
        t.Fatalf("want version=1, got %d", version)
    }
}

func TestMigratePragmas(t *testing.T) {
    dir := t.TempDir()
    path := filepath.Join(dir, "test.db")
    conn, _ := Open(path)
    defer conn.Close()
    _ = Migrate(conn)

    var jm string
    conn.QueryRow("PRAGMA journal_mode").Scan(&jm)
    if jm != "wal" {
        t.Errorf("journal_mode: want wal, got %s", jm)
    }
    var fk int
    conn.QueryRow("PRAGMA foreign_keys").Scan(&fk)
    if fk != 1 {
        t.Errorf("foreign_keys: want 1, got %d", fk)
    }
}

```



Step 3: Run test to verify FAIL

Run: `cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 go test -short -run Migrate ./internal/db/ -v` Expected: FAIL (Open / Migrate undefined).



Step 4: Implement internal/db/conn.go

```

package db

import (
    "database/sql"
    "fmt"

    _ "modernc.org/sqlite"
)

func Open(path string) (*sql.DB, error) {
    // modernc.org/sqlite uses _pragma= syntax for PRAGMA setup.
    dsn := fmt.Sprintf("file:%s?
        _pragma=journal_mode(WAL)&_pragma=foreign_keys(on)&_pragma=synchronous(NORMAL)
        path)
    conn, err := sql.Open("sqlite", dsn) // driver name "sqlite"
        (not "sqlite3") for modernc
    if err != nil {
        return nil, fmt.Errorf("sql.Open: %w", err)
    }
    conn.SetMaxOpenConns(1) // SQLite single-writer; reads also
        serialized for simplicity
    if err := conn.Ping(); err != nil {
        conn.Close()
        return nil, fmt.Errorf("ping: %w", err)
    }
    return conn, nil
}

```

Driver-swap note (post-Task-1 review): Task 1's commit ddcf5f7 swapped github.com/mattn/go-sqlite3 (CGO-only) for modernc.org/sqlite (pure Go). The blank import above locks the dep into the build graph so go mod tidy no longer strips it. Driver name is "sqlite" (not "sqlite3"); DSN uses _pragma=...(...) syntax. All subsequent tasks reference this driver.



Step 5: Implement internal/db/migrate.go

```

package db

import (
    "database/sql"
    "embed"
    "fmt"
    "sort"
    "strconv"
    "strings"
)

//go:embed migrations/*.sql
var migrationsFS embed.FS

```

```

func Migrate(conn *sql.DB) error {
    if _, err := conn.Exec(`CREATE TABLE IF NOT EXISTS
        schema_migrations (
            version INTEGER PRIMARY KEY,
            applied_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
        )`); err != nil {
        return fmt.Errorf("create schema_migrations: %w", err)
    }
    applied := map[int]bool{}
    rows, err := conn.Query("SELECT version FROM
        schema_migrations")
    if err != nil {
        return fmt.Errorf("query versions: %w", err)
    }
    for rows.Next() {
        var v int
        _ = rows.Scan(&v)
        applied[v] = true
    }
    rows.Close()

    entries, err := migrationsFS.ReadDir("migrations")
    if err != nil {
        return fmt.Errorf("read migrations dir: %w", err)
    }
    type mig struct {
        version int
        name     string
    }
    var migs []mig
    for _, e := range entries {
        name := e.Name()
        if !strings.HasSuffix(name, ".sql") {
            continue
        }
        verStr := strings.SplitN(name, "_", 2)[0]
        v, err := strconv.Atoi(verStr)
        if err != nil {
            return fmt.Errorf("bad migration name %s: %w", name,
                err)
        }
        migs = append(migs, mig{v, name})
    }
    sort.Slice(migs, func(i, j int) bool { return migs[i].version
        < migs[j].version })

    for _, m := range migs {
        if applied[m.version] {
            continue
        }
        body, err := migrationsFS.ReadFile("migrations/" + m.name)

```

```

    if err != nil {
        return fmt.Errorf("read %s: %w", m.name, err)
    }
    tx, err := conn.Begin()
    if err != nil {
        return fmt.Errorf("begin tx for %s: %w", m.name, err)
    }
    if _, err := tx.Exec(string(body)); err != nil {
        tx.Rollback()
        return fmt.Errorf("apply %s: %w", m.name, err)
    }
    // 001_init.sql does its own schema_migrations INSERT.
    // Subsequent migrations rely on this stepper to track.
    if m.version > 1 {
        if _, err := tx.Exec("INSERT INTO
schema_migrations(version) VALUES(?)", m.version); err !=
nil {
            tx.Rollback()
            return fmt.Errorf("record version %d: %w",
m.version, err)
        }
    }
    if err := tx.Commit(); err != nil {
        return fmt.Errorf("commit %s: %w", m.name, err)
    }
}
return nil
}
}

```



Step 6: Run tests to verify PASS

Run: `cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 go test -short -run Migrate ./internal/db/ -v` Expected: 3 PASS.



Step 7: Commit

```

git add qBitTorrent-go/internal/db/
git commit -m "$(cat <<'EOF'
feat(boba-jackett): SQLite schema 001_init + migrations stepper +
tests

```

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```

EOF
)"

```

Task 4: envfile parser

Files: - Create: qBitTorrent-go/internal/envfile/parse.go - Test: qBitTorrent-go/internal/envfile/parse_test.go



Step 1: Write failing test

```
package envfile
```

```
import (
    "reflect"
    "strings"
    "testing"
)

func TestParseAllForms(t *testing.T) {
    src := `# header comment
FOO=bar
QUOTED="hello world"
SINGLE='hi'
EMPTY=
SPACED = trimmed
# another comment
DUP=first
DUP=second
`

    got, err := Parse(strings.NewReader(src))
    if err != nil {
        t.Fatalf("Parse: %v", err)
    }
    want := map[string]string{
        "FOO": "bar", "QUOTED": "hello world", "SINGLE": "hi",
        "EMPTY": "", "SPACED": "trimmed", "DUP": "second",
    }
    if !reflect.DeepEqual(got, want) {
        t.Fatalf("got %+v want %+v", got, want)
    }
}

func TestParseHandlesBlanks(t *testing.T) {
    src := "\n\n\nFOO=bar\n\n"
    got, _ := Parse(strings.NewReader(src))
    if got["FOO"] != "bar" {
        t.Fatalf("got %+v", got)
    }
}
```



Step 2: Run — expect FAIL (Parse undefined)

Run: cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 go test
-short ./internal/envfile/ -v



Step 3: Implement parse.go

```
package envfile
```

```
import (  
    "bufio"  
    "io"  
    "strings"  
)  
  
// Parse returns the LAST occurrence for duplicate keys (matches  
    shell behavior).  
func Parse(r io.Reader) (map[string]string, error) {  
    out := map[string]string{}  
    sc := bufio.NewScanner(r)  
    for sc.Scan() {  
        line := sc.Text()  
        t := strings.TrimSpace(line)  
        if t == "" || strings.HasPrefix(t, "#") {  
            continue  
        }  
        eq := strings.IndexByte(t, '=')  
        if eq < 0 {  
            continue  
        }  
        k := strings.TrimSpace(t[:eq])  
        v := strings.TrimSpace(t[eq+1:])  
        // strip matching surrounding quotes  
        if len(v) >= 2 {  
            if (v[0] == '"' && v[len(v)-1] == '"') || (v[0] ==  
                '\'' && v[len(v)-1] == '\') {  
                v = v[1 : len(v)-1]  
            }  
        }  
        out[k] = v  
    }  
    if err := sc.Err(); err != nil {  
        return nil, err  
    }  
    return out, nil  
}
```



Step 4: Run — expect 2 PASS

```
Run: cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 go test
-short ./internal/envfile/ -v
```



Step 5: Commit

```
git add qBitTorrent-go/internal/envfile/parse.go qBitTorrent-go/
      internal/envfile/parse_test.go
git commit -m "$(cat <<'EOF'
feat(boba-jackett): envfile parser handles comments/quotes/blanks/
      dups
```

```
Co-Authored-By: Claude Opus 4.7 (1M context)
      <noreply@anthropic.com>
```

```
EOF
)"
```

Task 5: envfile atomic writer

Files: - Create: qBitTorrent-go/internal/envfile/write.go - Test:
qBitTorrent-go/internal/envfile/write_test.go



Step 1: Write failing tests

```
package envfile
```

```
import (
    "os"
    "path/filepath"
    "strings"
    "sync"
    "testing"
)
```

```
func TestWriteRoundTrip(t *testing.T) {
    dir := t.TempDir()
    p := filepath.Join(dir, ".env")
    os.WriteFile(p, []byte("F00=bar\n"), 0600)
    if err := Upsert(p, map[string]string{"BAZ": "qux", "F00":
        "baz"}); err != nil {
        t.Fatalf("Upsert: %v", err)
    }
    body, _ := os.ReadFile(p)
    s := string(body)
    if !strings.Contains(s, "BAZ=qux") {
        t.Fatalf("missing BAZ=qux:\n%s", s)
    }
    if !strings.Contains(s, "F00=baz") {
        t.Fatalf("missing F00=baz:\n%s", s)
    }
}
```

```

    }
    if strings.Contains(s, "F00=bar") {
        t.Fatalf("old F00=bar still present:\n%s", s)
    }
}

func TestWriteMode0600(t *testing.T) {
    dir := t.TempDir()
    p := filepath.Join(dir, ".env")
    os.WriteFile(p, []byte("F00=bar\n"), 0644)
    _ = Upsert(p, map[string]string{"F00": "baz"})
    st, _ := os.Stat(p)
    if st.Mode().Perm() != 0600 {
        t.Fatalf("want mode 0600, got %v", st.Mode().Perm())
    }
}

func TestDeleteKey(t *testing.T) {
    dir := t.TempDir()
    p := filepath.Join(dir, ".env")
    os.WriteFile(p, []byte("F00=bar\nBAZ=qux\n"), 0600)
    if err := Delete(p, []string{"F00"}); err != nil {
        t.Fatalf("Delete: %v", err)
    }
    body, _ := os.ReadFile(p)
    if strings.Contains(string(body), "F00=") {
        t.Fatalf("F00 not deleted:\n%s", body)
    }
    if !strings.Contains(string(body), "BAZ=qux") {
        t.Fatalf("BAZ removed unexpectedly:\n%s", body)
    }
}

func TestConcurrentWritesNoCorruption(t *testing.T) {
    dir := t.TempDir()
    p := filepath.Join(dir, ".env")
    os.WriteFile(p, []byte(""), 0600)
    var wg sync.WaitGroup
    for i := 0; i < 50; i++ {
        i := i
        wg.Add(1)
        go func() {
            defer wg.Done()
            _ = Upsert(p, map[string]string{"K": "v"})
            _ = Upsert(p, map[string]string{"K2": "v2"})
            _ = i
        }()
    }
    wg.Wait()
    body, _ := os.ReadFile(p)
    got, err := Parse(strings.NewReader(string(body)))
    if err != nil {

```

```

        t.Fatalf("post-concurrent parse: %v\nbody:\n%s", err,
            body)
    }
    if got["K"] != "v" || got["K2"] != "v2" {
        t.Fatalf("post-concurrent values lost: %+v", got)
    }
}

```



Step 2: Run — expect FAIL



Step 3: Implement write.go

```
package envfile
```

```

import (
    "fmt"
    "os"
    "path/filepath"
    "sort"
    "strings"
    "sync"
)

var writerMu sync.Mutex

func Upsert(path string, kv map[string]string) error {
    writerMu.Lock()
    defer writerMu.Unlock()
    return mutate(path, func(lines []string) []string {
        seen := map[string]bool{}
        out := make([]string, 0, len(lines)+len(kv))
        for _, l := range lines {
            t := strings.TrimSpace(l)
            if t == "" || strings.HasPrefix(t, "#") {
                out = append(out, l)
                continue
            }
            eq := strings.IndexByte(t, '=')
            if eq < 0 {
                out = append(out, l)
                continue
            }
            k := strings.TrimSpace(t[:eq])
            if v, ok := kv[k]; ok {
                if seen[k] {
                    continue // drop duplicates
                }
                out = append(out, fmt.Sprintf("%s=%s", k, v))
                seen[k] = true
            }
        }
        return out
    })
}

```

```

        continue
    }
    out = append(out, l)
}
// append new keys (sorted for determinism)
newKeys := make([]string, 0)
for k := range kv {
    if !seen[k] {
        newKeys = append(newKeys, k)
    }
}
sort.Strings(newKeys)
for _, k := range newKeys {
    out = append(out, fmt.Sprintf("%s=%s", k, kv[k]))
}
return out
})
}

```

```

func Delete(path string, keys []string) error {
    writerMu.Lock()
    defer writerMu.Unlock()
    drop := map[string]bool{}
    for _, k := range keys {
        drop[k] = true
    }
    return mutate(path, func(lines []string) []string {
        out := make([]string, 0, len(lines))
        for _, l := range lines {
            t := strings.TrimSpace(l)
            if t == "" || strings.HasPrefix(t, "#") {
                out = append(out, l)
                continue
            }
            eq := strings.IndexByte(t, '=')
            if eq < 0 {
                out = append(out, l)
                continue
            }
            k := strings.TrimSpace(t[:eq])
            if drop[k] {
                continue
            }
            out = append(out, l)
        }
        return out
    })
}

```

```

func mutate(path string, fn func([]string) []string) error {
    body, err := os.ReadFile(path)
    if err != nil && !os.IsNotExist(err) {

```

```

        return fmt.Errorf("read: %w", err)
    }
    lines := []string{}
    if len(body) > 0 {
        lines = strings.Split(strings.TrimRight(string(body),
            "\n"), "\n")
    }
    out := fn(lines)
    tmp := path + ".tmp"
    f, err := os.OpenFile(tmp, os.O_WRONLY|os.O_CREATE|
        os.O_TRUNC, 0600)
    if err != nil {
        return fmt.Errorf("open tmp: %w", err)
    }
    if _, err := f.WriteString(strings.Join(out, "\n") + "\n");
        err != nil {
        f.Close()
        os.Remove(tmp)
        return fmt.Errorf("write: %w", err)
    }
    if err := f.Sync(); err != nil {
        f.Close()
        os.Remove(tmp)
        return fmt.Errorf("fsync tmp: %w", err)
    }
    if err := f.Close(); err != nil {
        os.Remove(tmp)
        return fmt.Errorf("close tmp: %w", err)
    }
    if err := os.Chmod(tmp, 0600); err != nil {
        os.Remove(tmp)
        return fmt.Errorf("chmod tmp: %w", err)
    }
    if err := os.Rename(tmp, path); err != nil {
        os.Remove(tmp)
        return fmt.Errorf("rename: %w", err)
    }
    // fsync parent dir
    dir := filepath.Dir(path)
    d, err := os.Open(dir)
    if err == nil {
        _ = d.Sync()
        d.Close()
    }
    return nil
}

```



Step 4: Run — expect 4 PASS

```
Run: cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 go test
-short ./internal/envfile/ -v
```



Step 5: Commit

```
git add qBitTorrent-go/internal/envfile/write.go qBitTorrent-go/
      internal/envfile/write_test.go
git commit -m "$(cat <<'EOF'
feat(boba-jackett): atomic envfile Upsert/Delete with mutex +
      fsync
```

Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>

```
EOF
)"
```

Task 6: Credentials repo (encrypted CRUD)

Files: - Create: qBitTorrent-go/internal/db/repos/credentials.go -
Test: qBitTorrent-go/internal/db/repos/credentials_test.go



Step 1: Write failing tests

```
package repos
```

```
import (
    "crypto/rand"
    "path/filepath"
    "testing"
    "time"

    "github.com/milos85vasic/qBitTorrent-go/internal/db"
)

func freshConn(t *testing.T) *Credentials {
    t.Helper()
    dir := t.TempDir()
    conn, err := db.Open(filepath.Join(dir, "t.db"))
    if err != nil {
        t.Fatalf("Open: %v", err)
    }
    if err := db.Migrate(conn); err != nil {
        t.Fatalf("Migrate: %v", err)
    }
    key := make([]byte, 32)
    rand.Read(key)
    return NewCredentials(conn, key)
}
```

```

func TestUpsertAndGet(t *testing.T) {
    r := freshConn(t)
    if err := r.Upsert("RUTRACKER", "userpass", strPtr("u"),
        strPtr("p"), nil); err != nil {
        t.Fatalf("Upsert: %v", err)
    }
    got, err := r.Get("RUTRACKER")
    if err != nil {
        t.Fatalf("Get: %v", err)
    }
    if got.Name != "RUTRACKER" || got.Kind != "userpass" {
        t.Fatalf("bad row: %+v", got)
    }
    if got.Username != "u" || got.Password != "p" {
        t.Fatalf("decrypt mismatch: %+v", got)
    }
}

func TestList(t *testing.T) {
    r := freshConn(t)
    r.Upsert("RUTRACKER", "userpass", strPtr("u"), strPtr("p"),
        nil)
    r.Upsert("IPTORRENTS", "cookie", nil, nil, strPtr("c"))
    rows, err := r.List()
    if err != nil {
        t.Fatalf("List: %v", err)
    }
    if len(rows) != 2 {
        t.Fatalf("want 2 rows, got %d", len(rows))
    }
}

func TestDelete(t *testing.T) {
    r := freshConn(t)
    r.Upsert("X", "userpass", strPtr("u"), strPtr("p"), nil)
    if err := r.Delete("X"); err != nil {
        t.Fatalf("Delete: %v", err)
    }
    _, err := r.Get("X")
    if err == nil {
        t.Fatal("expected not-found")
    }
}

func TestUpsertOverwritesUpdatedAt(t *testing.T) {
    r := freshConn(t)
    r.Upsert("X", "userpass", strPtr("u"), strPtr("p"), nil)
    first, _ := r.Get("X")
    time.Sleep(1100 * time.Millisecond)
    r.Upsert("X", "userpass", strPtr("u2"), strPtr("p2"), nil)
    second, _ := r.Get("X")

```



```

    if !second.UpdatedAt.After(first.UpdatedAt) {
        t.Fatalf("updated_at did not advance: %v then %v",
            first.UpdatedAt, second.UpdatedAt)
    }
}

```

```

func strPtr(s string) *string { return &s }

```



Step 2: Run — expect FAIL



Step 3: Implement internal/db/repos/credentials.go

```

package repos

```

```

import (
    "database/sql"
    "errors"
    "fmt"
    "time"

    "github.com/milos85vasic/qBitTorrent-go/internal/db"
)

```

```

type Credential struct {
    Name          string
    Kind          string
    Username      string
    Password      string
    Cookies       string
    HasUsername   bool
    HasPassword   bool
    HasCookies    bool
    CreatedAt     time.Time
    UpdatedAt     time.Time
    LastUsedAt    *time.Time
}

```

```

type Credentials struct {
    conn *sql.DB
    key  []byte
}

```

```

var ErrNotFound = errors.New("repos: credential not found")

```

```

func NewCredentials(conn *sql.DB, key []byte) *Credentials {
    return &Credentials{conn: conn, key: key}
}

```

```

func (c *Credentials) Upsert(name, kind string, user, pass,
    cookies *string) error {

```

```

var (
    uEnc, pEnc, cEnc []byte
    err                error
)
if user != nil && *user != "" {
    uEnc, err = db.Encrypt(c.key, *user)
    if err != nil {
        return fmt.Errorf("encrypt username: %w", err)
    }
}
if pass != nil && *pass != "" {
    pEnc, err = db.Encrypt(c.key, *pass)
    if err != nil {
        return fmt.Errorf("encrypt password: %w", err)
    }
}
if cookies != nil && *cookies != "" {
    cEnc, err = db.Encrypt(c.key, *cookies)
    if err != nil {
        return fmt.Errorf("encrypt cookies: %w", err)
    }
}
now := time.Now().UTC()
_, err = c.conn.Exec(`
    INSERT INTO credentials(name, kind, username_enc,
        password_enc, cookies_enc, created_at, updated_at)
    VALUES(?, ?, ?, ?, ?, ?, ?)
    ON CONFLICT(name) DO UPDATE SET
        kind=excluded.kind,
        username_enc=COALESCE(excluded.username_enc,
            username_enc),
        password_enc=COALESCE(excluded.password_enc,
            password_enc),
        cookies_enc=COALESCE(excluded.cookies_enc,
            cookies_enc),
        updated_at=excluded.updated_at
`, name, kind, uEnc, pEnc, cEnc, now, now)
return err
}

func (c *Credentials) Get(name string) (*Credential, error) {
    row := c.conn.QueryRow(`SELECT name, kind, username_enc,
        password_enc, cookies_enc,
        created_at, updated_at, last_used_at FROM credentials
        WHERE name=?`, name)
    cr := &Credential{}
    var uEnc, pEnc, cEnc []byte
    if err := row.Scan(&cr.Name, &cr.Kind, &uEnc, &pEnc, &cEnc,
        &cr.CreatedAt, &cr.UpdatedAt, &cr.LastUsedAt); err != nil
    {
        if errors.Is(err, sql.ErrNoRows) {
            return nil, ErrNotFound
        }
    }
}

```

```

        return nil, fmt.Errorf("scan: %w", err)
    }
    if uEnc != nil {
        s, err := db.Decrypt(c.key, uEnc)
        if err != nil {
            return nil, fmt.Errorf("decrypt username: %w", err)
        }
        cr.Username = s
        cr.HasUsername = true
    }
    if pEnc != nil {
        s, err := db.Decrypt(c.key, pEnc)
        if err != nil {
            return nil, fmt.Errorf("decrypt password: %w", err)
        }
        cr.Password = s
        cr.HasPassword = true
    }
    if cEnc != nil {
        s, err := db.Decrypt(c.key, cEnc)
        if err != nil {
            return nil, fmt.Errorf("decrypt cookies: %w", err)
        }
        cr.Cookies = s
        cr.HasCookies = true
    }
    return cr, nil
}

func (c *Credentials) List() ([]*Credential, error) {
    rows, err := c.conn.Query(`SELECT name, kind, username_enc IS
        NOT NULL,
        password_enc IS NOT NULL, cookies_enc IS NOT NULL,
        created_at, updated_at, last_used_at FROM credentials
        ORDER BY name`)
    if err != nil {
        return nil, err
    }
    defer rows.Close()
    var out []*Credential
    for rows.Next() {
        cr := &Credential{}
        if err := rows.Scan(&cr.Name, &cr.Kind, &cr.HasUsername,
            &cr.HasPassword, &cr.HasCookies,
            &cr.CreatedAt, &cr.UpdatedAt, &cr.LastUsedAt); err !=
            nil {
            return nil, err
        }
        out = append(out, cr)
    }
    return out, rows.Err()
}

```

```

func (c *Credentials) Delete(name string) error {
    _, err := c.conn.Exec("DELETE FROM credentials WHERE name=?",
        name)
    return err
}

func (c *Credentials) MarkUsed(name string) error {
    _, err := c.conn.Exec("UPDATE credentials SET
        last_used_at=CURRENT_TIMESTAMP WHERE name=?", name)
    return err
}

```



Step 4: Run — expect 4 PASS

Run: `cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 go test -short ./internal/db/repos/ -run Credential -v`



Step 5: Commit

```

git add qBitTorrent-go/internal/db/repos/credentials.go
      qBitTorrent-go/internal/db/repos/credentials_test.go
git commit -m "$(cat <<'EOF'
feat(boba-jackett): credentials repo with encrypted CRUD + tests

Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>
EOF
)"

```

Task 7: Indexers repo

Files: - Create: `qBitTorrent-go/internal/db/repos/indexers.go` - Test: `qBitTorrent-go/internal/db/repos/indexers_test.go`



Step 1: Write failing tests

```
package repos
```

```

import (
    "path/filepath"
    "testing"

    "github.com/milos85vasic/qBitTorrent-go/internal/db"
)

func indexersConn(t *testing.T) *Indexers {

```

```

    t.Helper()
    dir := t.TempDir()
    conn, _ := db.Open(filepath.Join(dir, "t.db"))
    _ = db.Migrate(conn)
    return NewIndexers(conn)
}

func TestIndexersUpsertAndList(t *testing.T) {
    r := indexersConn(t)
    if err := r.Upsert(&Indexer{
        ID: "rutracker", DisplayName: "RuTracker.org", Type:
        "private",
        ConfiguredAtJackett: true, EnabledForSearch: true,
    }); err != nil {
        t.Fatalf("Upsert: %v", err)
    }
    rows, err := r.List()
    if err != nil {
        t.Fatalf("List: %v", err)
    }
    if len(rows) != 1 || rows[0].ID != "rutracker" {
        t.Fatalf("got %v", rows)
    }
}

func TestIndexersDelete(t *testing.T) {
    r := indexersConn(t)
    r.Upsert(&Indexer{ID: "x", DisplayName: "X", Type: "public",
        EnabledForSearch: true})
    if err := r.Delete("x"); err != nil {
        t.Fatalf("Delete: %v", err)
    }
    rows, _ := r.List()
    if len(rows) != 0 {
        t.Fatalf("want 0, got %d", len(rows))
    }
}

func TestIndexersToggleEnabled(t *testing.T) {
    r := indexersConn(t)
    r.Upsert(&Indexer{ID: "x", DisplayName: "X", Type: "public",
        EnabledForSearch: true})
    if err := r.SetEnabled("x", false); err != nil {
        t.Fatalf("SetEnabled: %v", err)
    }
    rows, _ := r.List()
    if rows[0].EnabledForSearch {
        t.Fatal("expected disabled")
    }
}

```



Step 2: Run — expect FAIL



Step 3: Implement `internal/db/repos/indexers.go`

```
package repos
```

```
import (  
    "database/sql"  
    "time"  
)
```

```
type Indexer struct {  
    ID                string  
    DisplayName       string  
    Type              string  
    ConfiguredAtJackett bool  
    LinkedCredentialName *string  
    EnabledForSearch   bool  
    LastJackettSyncAt  *time.Time  
    LastTestStatus     *string  
    LastTestAt         *time.Time  
}
```

```
type Indexers struct{ conn *sql.DB }
```

```
func NewIndexers(conn *sql.DB) *Indexers { return &Indexers{conn:  
    conn} }
```

```
func (r *Indexers) Upsert(i *Indexer) error {  
    _, err := r.conn.Exec(`  
        INSERT INTO indexers(id, display_name, type,  
            configured_at_jackett,  
                linked_credential_name, enabled_for_search,  
                    last_jackett_sync_at,  
                        last_test_status, last_test_at)  
        VALUES(?, ?, ?, ?, ?, ?, ?, ?, ?)  
        ON CONFLICT(id) DO UPDATE SET  
            display_name=excluded.display_name,  
            type=excluded.type,  
            configured_at_jackett=excluded.configured_at_jackett,  
  
            linked_credential_name=excluded.linked_credential_name,  
            enabled_for_search=excluded.enabled_for_search,  
            last_jackett_sync_at=excluded.last_jackett_sync_at,  
            last_test_status=excluded.last_test_status,  
            last_test_at=excluded.last_test_at  
    `, i.ID, i.DisplayName, i.Type, i.ConfiguredAtJackett,  
        i.LinkedCredentialName, i.EnabledForSearch,  
        i.LastJackettSyncAt,  
        i.LastTestStatus, i.LastTestAt)  
    return err
```

```

}

func (r *Indexers) List() ([]*Indexer, error) {
    rows, err := r.conn.Query(`SELECT id, display_name, type,
        configured_at_jackett,
        linked_credential_name, enabled_for_search,
        last_jackett_sync_at,
        last_test_status, last_test_at FROM indexers ORDER BY id`)
    if err != nil {
        return nil, err
    }
    defer rows.Close()
    var out []*Indexer
    for rows.Next() {
        i := &Indexer{}
        if err := rows.Scan(&i.ID, &i.DisplayName, &i.Type,
            &i.ConfiguredAtJackett,
            &i.LinkedCredentialName, &i.EnabledForSearch,
            &i.LastJackettSyncAt,
            &i.LastTestStatus, &i.LastTestAt); err != nil {
            return nil, err
        }
        out = append(out, i)
    }
    return out, rows.Err()
}

func (r *Indexers) Delete(id string) error {
    _, err := r.conn.Exec("DELETE FROM indexers WHERE id=?", id)
    return err
}

func (r *Indexers) SetEnabled(id string, enabled bool) error {
    _, err := r.conn.Exec("UPDATE indexers SET
        enabled_for_search=? WHERE id=?", enabled, id)
    return err
}

func (r *Indexers) RecordTest(id, status string) error {
    now := time.Now().UTC()
    _, err :=
        r.conn.Exec("UPDATE indexers SET last_test_status=?,
            last_test_at=? WHERE id=?", status, now, id)
    return err
}

```



Step 4: Run — expect 3 PASS



Step 5: Commit

```
git add qBitTorrent-go/internal/db/repos/indexers.go qBitTorrent-
      go/internal/db/repos/indexers_test.go
git commit -m "$(cat <<'EOF'
feat(boba-jackett): indexers repo CRUD + enable toggle + tests

Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>

EOF
)"
```

Task 8: Catalog cache repo

Files: - Create: qBitTorrent-go/internal/db/repos/catalog.go - Test: qBitTorrent-go/internal/db/repos/catalog_test.go



Step 1: Write failing tests — minimal CRUD: Upsert, Get, Query(filter), OlderThan(d) []ID. Tests follow the same pattern as Task 7.



Step 2: Run — expect FAIL



Step 3: Implement catalog.go with fields matching catalog_cache schema. Query supports search (LIKE on display_name+description), type=, language=, paginated LIMIT ?, OFFSET ?. Use prepared statements.



Step 4: Run — expect PASS



Step 5: Commit: feat(boba-jackett): catalog_cache repo with paginated query

Task 9: Autoconfig runs repo + overrides repo

Two related repos in one task — both are simple CRUD.

Files: - Create: qBitTorrent-go/internal/db/repos/runs.go - Test: qBitTorrent-go/internal/db/repos/runs_test.go - Create: qBitTorrent-go/internal/db/repos/overrides.go - Test: qBitTorrent-go/internal/db/repos/overrides_test.go



Step 1: Tests for runs: `Insert(result), List(limit), Get(id)`
returning summary blob.

☐

Step 2: Tests for overrides: `List(), Upsert(env_name, indexer_id),`
`Delete(env_name).`

☐

Step 3: Run — expect FAIL

☐

**Step 4: Implement both files following the same `*sql.DB`
constructor pattern as Tasks 6/7.**

☐

Step 5: Run — expect PASS

☐

Step 6: Commit:

`feat(boba-jackett): autoconfig_runs + overrides repos`

Phase 2 — Bootstrap & Jackett port (Tasks 10-14)

Task 10: Bootstrap (master key autogen + import)

Files: - Create: `qBittorrent-go/internal/bootstrap/bootstrap.go` -
Create: `qBittorrent-go/internal/bootstrap/import.go` - Test:
`qBittorrent-go/internal/bootstrap/bootstrap_test.go`

☐

Step 1: Write failing tests

```
package bootstrap
```

```
import (  
    "os"  
    "path/filepath"  
    "strings"  
    "testing"  
  
    "github.com/milos85vasic/qBittorrent-go/internal/envfile"  
)
```

```
func TestEnsureMasterKeyGeneratesIfMissing(t *testing.T) {  
    dir := t.TempDir()  
    envP := filepath.Join(dir, ".env")  
    os.WriteFile(envP, []byte("F00=bar\n"), 0600)
```

```

key, generated, err := EnsureMasterKey(envP)
if err != nil {
    t.Fatalf("EnsureMasterKey: %v", err)
}
if !generated {
    t.Fatal("expected generated=true")
}
if len(key) != 32 {
    t.Fatalf("want 32 bytes, got %d", len(key))
}
body, _ := os.ReadFile(envP)
if !strings.Contains(string(body), "BOBA_MASTER_KEY=") {
    t.Fatalf("key not persisted to .env:\n%s", body)
}
if !strings.Contains(string(body), "DO NOT LOSE") {
    t.Fatalf("warning header missing:\n%s", body)
}
}

func TestEnsureMasterKeyIdempotent(t *testing.T) {
    dir := t.TempDir()
    envP := filepath.Join(dir, ".env")
    os.WriteFile(envP,
        []byte("BOBA_MASTER_KEY=00112233445566778899aabbccddeeff00112233445566778800000000"),
        0600)
    _, generated, _ := EnsureMasterKey(envP)
    if generated {
        t.Fatal("expected generated=false")
    }
}

func TestImportFromEnvBundlesTriples(t *testing.T) {
    src, _ :=
        envfile.Parse(strings.NewReader(`RUTRACKER_USERNAME=u
RUTRACKER_PASSWORD=p
IPTORRENTS_COOKIES=c
INCOMPLETE_USERNAME=onlyuser
JACKETT_API_KEY=ignored
`))
    bundles := DiscoverCredentialBundles(src, defaultExclude())
    if len(bundles) != 2 {
        t.Fatalf("want 2 (RUTRACKER, IPTORRENTS), got %+v",
            bundles)
    }
}

```



Step 2: Run — expect FAIL



Step 3: Implement bootstrap.go

```

package bootstrap

import (
    "crypto/rand"
    "encoding/hex"
    "fmt"
    "os"

    "github.com/milos85vasic/qBitTorrent-go/internal/envfile"
)

const masterKeyHeader = `# === BOBA SYSTEM ===
# Master key for credential encryption-at-rest in config/boba.db.
# DO NOT LOSE THIS – credentials become unrecoverable without it.
# To rotate: see docs/BOBA_DATABASE.md § "Key Rotation".`

func EnsureMasterKey(envPath string) (key []byte, generated bool,
    err error) {
    f, err := os.Open(envPath)
    if err != nil {
        return nil, false, fmt.Errorf("open .env: %w", err)
    }
    parsed, err := envfile.Parse(f)
    f.Close()
    if err != nil {
        return nil, false, fmt.Errorf("parse .env: %w", err)
    }
    if existing, ok := parsed["BOBA_MASTER_KEY"]; ok &&
        len(existing) == 64 {
        k, err := hex.DecodeString(existing)
        if err == nil && len(k) == 32 {
            return k, false, nil
        }
    }
    raw := make([]byte, 32)
    if _, err := rand.Read(raw); err != nil {
        return nil, false, fmt.Errorf("rand.Read: %w", err)
    }
    hexKey := hex.EncodeToString(raw)
    // Append header (as comments) then key
    body, err := os.ReadFile(envPath)
    if err != nil {
        return nil, false, fmt.Errorf("read .env: %w", err)
    }
    out := string(body)
    if len(out) > 0 && out[len(out)-1] != '\n' {
        out += "\n"
    }
    out += "\n" + masterKeyHeader + "\n"
    if err := os.WriteFile(envPath+".tmp", []byte(out), 0600);
        err != nil {
        return nil, false, fmt.Errorf("write tmp: %w", err)
    }
}

```

```

    }
    if err := os.Rename(envPath+".tmp", envPath); err != nil {
        return nil, false, fmt.Errorf("rename: %w", err)
    }
    if err := envfile.Upsert(envPath,
        map[string]string{"BOBA_MASTER_KEY": hexKey}); err != nil {
        {
            return nil, false, fmt.Errorf("upsert key: %w", err)
        }
    }
    return raw, true, nil
}

var defaultExcludeSet = map[string]bool{
    "QBITTORRENT": true, "JACKETT": true, "WEBUI": true,
    "PROXY": true, "MERGE": true, "BRIDGE": true, "BOBA": true,
}

func defaultExclude() map[string]bool {
    out := make(map[string]bool, len(defaultExcludeSet))
    for k, v := range defaultExcludeSet {
        out[k] = v
    }
    return out
}

```



Step 4: Implement import.go

```

package bootstrap

import "regexp"

var credRE = regexp.MustCompile(`^([A-Z][A-Z0-9_]+?)_(USERNAME|
    PASSWORD|COOKIES)$`)

type CredBundle struct {
    Name      string
    Username  string
    Password  string
    Cookies   string
}

func DiscoverCredentialBundles(env map[string]string, exclude
    map[string]bool) []*CredBundle {
    groups := map[string]*CredBundle{}
    for k, v := range env {
        m := credRE.FindStringSubmatch(k)
        if m == nil {
            continue
        }
        name, kind := m[1], m[2]
        if exclude[name] {

```

```

        continue
    }
    b := groups[name]
    if b == nil {
        b = &CredBundle{Name: name}
        groups[name] = b
    }
    switch kind {
    case "USERNAME":
        b.Username = v
    case "PASSWORD":
        b.Password = v
    case "COOKIES":
        b.Cookies = v
    }
}
}
var out []*CredBundle
for _, b := range groups {
    hasUP := b.Username != "" && b.Password != ""
    hasC := b.Cookies != ""
    if hasUP || hasC {
        out = append(out, b)
    }
}
return out
}
}

```



Step 5: Run — expect 3 PASS



Step 6: Commit: feat(boba-jackett): bootstrap (master key
autogen + .env import)

Task 11: Jackett HTTP client (warmup + catalog)

Files: - Create: qBitTorrent-go/internal/jackett/client.go - Test:
qBitTorrent-go/internal/jackett/client_test.go



Step 1: Write failing tests using httpptest.NewServer

package jackett

```

import (
    "net/http"
    "net/http/httpptest"
    "strings"
    "testing"

```

)

```
func TestSessionWarmupAndCatalog(t *testing.T) {
    warmedUp := false
    srv := httptest.NewServer(http.HandlerFunc(func(w
        http.ResponseWriter, r *http.Request) {
            switch {
            case r.Method == "POST" && r.URL.Path == "/UI/Dashboard":
                http.SetCookie(w, &http.Cookie{Name: "Jackett",
                    Value: "session"})
                warmedUp = true
                w.WriteHeader(302)
            case r.URL.Path == "/api/v2.0/indexers":
                if !warmedUp {
                    w.WriteHeader(401)
                    return
                }
                w.Header().Set("Content-Type", "application/json")

                w.Write([]byte(`[{"id":"rutracker","name":"RuTracker.org","type":"private"
            }
        })))
    defer srv.Close()
    c := NewClient(srv.URL, "test-key")
    if err := c.WarmUp(); err != nil {
        t.Fatalf("WarmUp: %v", err)
    }
    cat, err := c.GetCatalog()
    if err != nil {
        t.Fatalf("GetCatalog: %v", err)
    }
    if len(cat) != 1 || cat[0].ID != "rutracker" {
        t.Fatalf("got %+v", cat)
    }
}
```

```
func TestPostConfig(t *testing.T) {
    var captured string
    srv := httptest.NewServer(http.HandlerFunc(func(w
        http.ResponseWriter, r *http.Request) {
            if r.Method == "POST" && strings.Contains(r.URL.Path, "/"
                config) {
                body := make([]byte, 1024)
                n, _ := r.Body.Read(body)
                captured = string(body[:n])
                w.WriteHeader(200)
            }
        })))
    defer srv.Close()
    c := NewClient(srv.URL, "k")
    body := []map[string]any{{"id": "username", "value": "u"}}
    if err := c.PostIndexerConfig("x", body); err != nil {
        t.Fatalf("Post: %v", err)
    }
}
```

```

    }
    if !strings.Contains(captured, `"username"`) {
        t.Fatalf("body not posted: %s", captured)
    }
}

```



Step 2: Run — expect FAIL



Step 3: Implement client.go

```

package jackett

import (
    "bytes"
    "encoding/json"
    "fmt"
    "net/http"
    "net/http/cookiejar"
    "net/url"
    "strings"
    "time"
)

type CatalogEntry struct {
    ID          string    `json:"id"`
    Name        string    `json:"name"`
    Type        string    `json:"type"`
    Configured  bool      `json:"configured"`
    Language    string    `json:"language"`
    Description string    `json:"description"`
}

type Client struct {
    base      string
    apiKey    string
    http      *http.Client
}

func NewClient(base, apiKey string) *Client {
    jar, _ := cookiejar.New(nil)
    return &Client{
        base:    strings.TrimRight(base, "/"),
        apiKey:  apiKey,
        http:    &http.Client{Timeout: 30 * time.Second, Jar: jar},
    }
}

func (c *Client) WarmUp() error {
    form := url.Values{"password": {""}}

```

```

req, _ := http.NewRequest("POST", c.base+"/UI/Dashboard",
    strings.NewReader(form.Encode()))
req.Header.Set("Content-Type", "application/x-www-form-
    urlencoded")
resp, err := c.http.Do(req)
if err != nil {
    return fmt.Errorf("warmup: %w", err)
}
defer resp.Body.Close()
return nil
}

func (c *Client) GetCatalog() ([]CatalogEntry, error) {
    u, _ := url.Parse(c.base + "/api/v2.0/indexers")
    q := u.Query()
    q.Set("apikey", c.apiKey)
    q.Set("configured", "false")
    u.RawQuery = q.Encode()
    req, _ := http.NewRequest("GET", u.String(), nil)
    req.Header.Set("Accept", "application/json")
    resp, err := c.http.Do(req)
    if err != nil {
        return nil, fmt.Errorf("get catalog: %w", err)
    }
    defer resp.Body.Close()
    if resp.StatusCode == 401 {
        return nil, fmt.Errorf("jackett_auth_failed")
    }
    if resp.StatusCode >= 400 {
        return nil, fmt.Errorf("jackett_catalog_http_%d",
            resp.StatusCode)
    }
    var entries []CatalogEntry
    if err := json.NewDecoder(resp.Body).Decode(&entries); err !=
        nil {
        return nil, fmt.Errorf("decode: %w", err)
    }
    return entries, nil
}

func (c *Client) GetIndexerTemplate(id string) ([]map[string]any,
    error) {
    u := fmt.Sprintf("%s/api/v2.0/indexers/%s/config?apikey=%s",
        c.base, id, c.apiKey)
    req, _ := http.NewRequest("GET", u, nil)
    req.Header.Set("Accept", "application/json")
    resp, err := c.http.Do(req)
    if err != nil {
        return nil, err
    }
    defer resp.Body.Close()
    if resp.StatusCode >= 400 {

```



```

        return nil, fmt.Errorf("template fetch HTTP %d",
            resp.StatusCode)
    }
    var raw any
    if err := json.NewDecoder(resp.Body).Decode(&raw); err != nil {
        {
            return nil, err
        }
    }
    switch v := raw.(type) {
    case []any:
        out := make([]map[string]any, 0, len(v))
        for _, item := range v {
            if m, ok := item.(map[string]any); ok {
                out = append(out, m)
            }
        }
        return out, nil
    case map[string]any:
        if cfg, ok := v["config"].([]any); ok {
            out := make([]map[string]any, 0, len(cfg))
            for _, item := range cfg {
                if m, ok := item.(map[string]any); ok {
                    out = append(out, m)
                }
            }
            return out, nil
        }
    }
    return nil, fmt.Errorf("unexpected template shape")
}

func (c *Client) PostIndexerConfig(id string, fields
    []map[string]any) error {
    u := fmt.Sprintf("%s/api/v2.0/indexers/%s/config?apikey=%s",
        c.base, id, c.apiKey)
    body, _ := json.Marshal(fields)
    req, _ := http.NewRequest("POST", u, bytes.NewReader(body))
    req.Header.Set("Content-Type", "application/json")
    resp, err := c.http.Do(req)
    if err != nil {
        return err
    }
    defer resp.Body.Close()
    if resp.StatusCode >= 400 {
        return fmt.Errorf("config POST HTTP %d", resp.StatusCode)
    }
    return nil
}

func (c *Client) DeleteIndexer(id string) error {
    u := fmt.Sprintf("%s/api/v2.0/indexers/%s?apikey=%s", c.base,
        id, c.apiKey)
    req, _ := http.NewRequest("DELETE", u, nil)

```

```

    resp, err := c.http.Do(req)
    if err != nil {
        return err
    }
    defer resp.Body.Close()
    if resp.StatusCode >= 400 && resp.StatusCode != 404 {
        return fmt.Errorf("delete HTTP %d", resp.StatusCode)
    }
    return nil
}

```



Step 4: Run — expect 2 PASS



Step 5: Commit:

feat(boba-jackett): Jackett HTTP client (warmup + catalog + config + delete)

Task 12: Matcher (port from Python)

Files: - Create: qBitTorrent-go/internal/jackett/matcher.go - Test: qBitTorrent-go/internal/jackett/matcher_test.go



Step 1: Port Python tests verbatim from tests/unit/merge_service/test_jackett_autoconfig.py — find all `_match_indexers` cases, translate to Go test functions covering: exact override, fuzzy match above threshold, ambiguous match, no match.



Step 2: Run — expect FAIL



Step 3: Implement matcher

Add github.com/agnivade/levenshtein dep:

```
cd qBitTorrent-go && go get github.com/agnivade/levenshtein
```

```
package jackett
```

```

import (
    "sort"
    "strings"

    "github.com/agnivade/levenshtein"
)

```

```

const FuzzyThreshold = 0.85

type AmbiguousMatch struct {
    EnvName    string
    Candidates []string
}

func ratio(a, b string) float64 {
    if a == "" && b == "" {
        return 1.0
    }
    d := levenshtein.ComputeDistance(a, b)
    maxLen := len(a)
    if len(b) > maxLen {
        maxLen = len(b)
    }
    if maxLen == 0 {
        return 1.0
    }
    return 1.0 - float64(d)/float64(maxLen)
}

func MatchIndexers(envNames []string, catalog []CatalogEntry,
    override map[string]string) (matched map[string]string,
    ambiguous []AmbiguousMatch, unmatched []string) {
    matched = map[string]string{}
    ids := map[string]bool{}
    for _, e := range catalog {
        ids[e.ID] = true
    }
    for _, name := range envNames {
        if t, ok := override[name]; ok && ids[t] {
            matched[name] = t
            continue
        }
        needle := strings.ToLower(name)
        type scored struct {
            id    string
            score float64
        }
        var scoredAll []scored
        for _, e := range catalog {
            s := ratio(needle, strings.ToLower(e.ID))
            if n := ratio(needle, strings.ToLower(e.Name)); n > s
            {
                s = n
            }
            if s >= FuzzyThreshold {
                scoredAll = append(scoredAll, scored{e.ID, s})
            }
        }
        if len(scoredAll) == 0 {

```

```

        unmatched = append(unmatched, name)
        continue
    }
    sort.Slice(scoredAll, func(i, j int) bool {
        if scoredAll[i].score != scoredAll[j].score {
            return scoredAll[i].score > scoredAll[j].score
        }
        return scoredAll[i].id < scoredAll[j].id
    })
    topScore := scoredAll[0].score
    var ties []string
    for _, s := range scoredAll {
        if s.score == topScore {
            ties = append(ties, s.id)
        }
    }
    if len(ties) == 1 {
        matched[name] = ties[0]
    } else {
        ambiguous = append(ambiguous, AmbiguousMatch{EnvName:
            name, Candidates: ties})
    }
}
return
}

```

☐

Step 4: Run — expect all PASS (matching Python parity).

☐

Step 5: Commit: feat(boba-jackett): matcher port from Python
(fuzzy + override)

Task 13: Autoconfig orchestrator (DB-aware)

Files: - Create: qBitTorrent-go/internal/jackett/autoconfig.go - Test:
qBitTorrent-go/internal/jackett/autoconfig_test.go

☐

Step 1: Write failing tests covering: no credentials, missing API
key, indexer requires cookie but cred has only userpass (skip with `n
o_compatible_credential_fields_for_indexer`), happy path with
mocked Jackett.

☐

Step 2: Run — expect FAIL

☐

Step 3: Implement orchestrator using credentials + indexers + runs repos. The orchestrator:

1. Loads all credential bundles from `credentials` repo.
2. Loads override map from `overrides` repo merged with `JACKETT_INDEXER_MAP` env.
3. WarmUp + GetCatalog via `jackett.Client`.
4. Calls `MatchIndexers`.
5. For each match: `GetIndexerTemplate` → fill fields → `PostIndexerConfig` → record indexer row.
6. Inserts run summary into `autoconfig_runs`.
7. Returns `AutoconfigResult` matching the Python shape.



Step 4: Run — expect PASS



Step 5: Commit: feat(boba-jackett): autoconfig orchestrator (DB-aware port)

Task 14: Logging redactor

Files: - Create: `qBittorrent-go/internal/logging/redactor.go` - Test: `qBittorrent-go/internal/logging/redactor_test.go`



Step 1: Write failing test

```
package logging
```

```
import (  
    "bytes"  
    "strings"  
    "testing"  
)
```

```
func TestRedactorReplacesSecrets(t *testing.T) {  
    var buf bytes.Buffer  
    r := NewRedactor(&buf)  
    r.AddSecret("supersecret123")  
    r.Write([]byte("user logged in with password=supersecret123  
        ok"))  
    if strings.Contains(buf.String(), "supersecret123") {  
        t.Fatalf("secret leaked: %s", buf.String())  
    }  
    if !strings.Contains(buf.String(), "****") {  
        t.Fatalf("no redaction marker: %s", buf.String())  
    }  
}
```

```

    }
}

func TestRedactorMultipleSecrets(t *testing.T) {
    var buf bytes.Buffer
    r := NewRedactor(&buf)
    r.AddSecret("aaa")
    r.AddSecret("bbb")
    r.Write([]byte("aaa bbb ccc"))
    if strings.Contains(buf.String(), "aaa") ||
        strings.Contains(buf.String(), "bbb") {
        t.Fatalf("secret leaked: %s", buf.String())
    }
}

```



Step 2: Run — expect FAIL



Step 3: Implement

```

package logging

import (
    "bytes"
    "io"
    "sync"
)

type Redactor struct {
    dest    io.Writer
    mu      sync.RWMutex
    secrets [][]byte
    mask    []byte
}

func NewRedactor(dest io.Writer) *Redactor {
    return &Redactor{dest: dest, mask: []byte("****")}
}

func (r *Redactor) AddSecret(s string) {
    if s == "" {
        return
    }
    r.mu.Lock()
    defer r.mu.Unlock()
    r.secrets = append(r.secrets, []byte(s))
}

func (r *Redactor) RemoveSecret(s string) {
    if s == "" {

```

```

        return
    }
    r.mu.Lock()
    defer r.mu.Unlock()
    target := []byte(s)
    for i, sec := range r.secrets {
        if bytes.Equal(sec, target) {
            r.secrets = append(r.secrets[:i], r.secrets[i+1:]...)
            return
        }
    }
}

func (r *Redactor) Write(p []byte) (int, error) {
    r.mu.RLock()
    defer r.mu.RUnlock()
    out := p
    for _, sec := range r.secrets {
        out = bytes.ReplaceAll(out, sec, r.mask)
    }
    _, err := r.dest.Write(out)
    return len(p), err
}

```



Step 4: Run — expect 2 PASS



Step 5: Commit: feat(boba-jackett): logging redactor for credential leak prevention

Phase 3 — HTTP API (Tasks 15-22)

Task 15: Auth middleware (admin/admin session)

Files: - Create: qBitTorrent-go/internal/jackettapi/auth_middleware.go - Test: qBitTorrent-go/internal/jackettapi/auth_middleware_test.go



Step 1: Write failing test that asserts: GET requests pass without auth, mutating requests (POST/PATCH/DELETE) without admin/admin Basic auth header → 401.



Step 2: Run — expect FAIL



Step 3: Implement middleware that wraps `http.Handler`, checks Authorization: Basic `base64("admin:admin")` for non-GET methods. (Matches the project's hardcoded creds per CLAUDE.md.)



Step 4: Run — expect PASS



Step 5: Commit: `feat(boba-jackett): auth middleware (admin/admin Basic for mutations)`

Task 16: Credentials API endpoints

Files: - Create: `qBittorrent-go/internal/jackettapi/credentials.go` -
Test: `qBittorrent-go/internal/jackettapi/credentials_test.go`



Step 1: Write failing tests for each endpoint in spec §8.1 using `httptest.NewServer` over the Credentials repo. Verify GET response shape NEVER contains plaintext values; POST adds row; DELETE removes; PATCH semantics.



Step 2: Run — expect FAIL



Step 3: Implement handlers with the JSON shapes from §8.1. Each handler also writes to `.env` via `envfile.Upsert/Delete` AFTER the DB write succeeds. Failure of `.env` write logs warning but doesn't fail the request (DB is canonical).

Wait — the spec demands hybrid C: BOTH writes must succeed. Re-read spec §7. Actually §7 says dashboard writes encrypted and atomic mirror — both required. So if `.env` write fails, we MUST roll back the DB. Implement transactional pattern:

1. Begin tx
2. DB upsert
3. `.env` upsert (atomic – file-level)
4. If `.env` fails: tx rollback + return 500
5. tx commit

Also: trigger autoconfig replay for that single tracker after success (spec §7 row 1).



Step 4: Run — expect PASS



Step 5: Commit:

feat(boba-jackett): credentials API with hybrid DB+env write

Task 17: Indexers API endpoints

Files: - Create: qBitTorrent-go/internal/jackettapi/indexers.go - Test: qBitTorrent-go/internal/jackettapi/indexers_test.go

☐

Step 1: Write failing tests for spec §8.2 endpoints: list, configure (POST), delete, test, toggle.

☐

Step 2: Run — expect FAIL

☐

Step 3: Implement handlers. POST /indexers/{id} fetches the template from Jackett, populates with credential's username/password/cookies (via Credentials repo Get), POSTs to Jackett, upserts the indexers row. DELETE removes from Jackett + DB.

☐

Step 4: Run — expect PASS

☐

Step 5: Commit: feat(boba-jackett): indexers API (configure/remove/test/toggle)

Task 18: Catalog API endpoints

Files: - Create: qBitTorrent-go/internal/jackettapi/catalog.go - Test: qBitTorrent-go/internal/jackettapi/catalog_test.go

☐

Step 1: Write failing tests for paginated query + force-refresh.

☐

Step 2: Run — expect FAIL

☐

Step 3: Implement GET /catalog with pagination + filters; POST /catalog/refresh re-fetches from Jackett and upserts all 620 rows.

☐

Step 4: Run — expect PASS

☐

Step 5: Commit: feat(boba-jackett): catalog API (paginated browse + refresh)

Task 19: Autoconfig runs API + manual trigger

Files: - Create: qBitTorrent-go/internal/jackettapi/runs.go - Test: qBitTorrent-go/internal/jackettapi/runs_test.go

☐

Step 1: Write failing tests for list/get/run.

☐

Step 2: Run — expect FAIL

☐

Step 3: Implement GET /autoconfig/runs (list with limit), GET /autoconfig/runs/{id}, POST /autoconfig/run (synchronous run).

☐

Step 4: Run — expect PASS

☐

Step 5: Commit: feat(boba-jackett): autoconfig runs history + manual trigger API

Task 20: Overrides API + Health endpoint

Files: - Create: qBitTorrent-go/internal/jackettapi/overrides.go - Create: qBitTorrent-go/internal/jackettapi/health.go - Test: qBitTorrent-go/internal/jackettapi/overrides_test.go - Test: qBitTorrent-go/internal/jackettapi/health_test.go

☐

Step 1: Write failing tests — overrides CRUD + healthz returns {status, db_ok, jackett_ok, version, uptime_s}.

☐

Step 2: Run — expect FAIL

☐

Step 3: Implement both. Health pings DB (SELECT 1) and Jackett (GET /health).

☐

Step 4: Run — expect PASS

☐

Step 5: Commit: feat(boba-jackett): overrides API + healthz

Task 21: Router + redactor wiring + main.go

Files: - Create: qBitTorrent-go/internal/jackettapi/router.go -
Modify: qBitTorrent-go/cmd/boba-jackett/main.go



Step 1: Implement router

```
package jackettapi

import (
    "net/http"
)

type Deps struct {
    // repos, clients, autoconfig orchestrator, redactor
}

func NewMux(d *Deps) http.Handler {
    mux := http.NewServeMux()
    // register all handlers from Tasks 16-20
    return WithAuth(d, mux)
}
```



Step 2: Wire cmd/boba-jackett/main.go to:

1. Read env (BOBA_DB_PATH, BOBA_ENV_PATH, JACKETT_URL, JACKETT_API_KEY, PORT).
2. EnsureMasterKey.
3. Open DB + Migrate.
4. Construct repos, Jackett client, redactor (set as os.Stdout/Stderr writer; populate from credentials.List() decrypted values once on boot).
5. Run first-pass autoconfig.
6. Start HTTP server on PORT.



Step 3: Build + run smoke

```
cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 bash scripts/build.sh
BOBA_DB_PATH=/tmp/test-boba.db BOBA_ENV_PATH=/tmp/test.env
JACKETT_URL=http://localhost:9117 JACKETT_API_KEY=$(jq -r
'.APIKey' ../config/jackett/Jackett/ServerConfig.json)
PORT=7189 ./bin/boba-jackett &

sleep 3
curl -sf http://localhost:7189/healthz && echo OK
kill %1
```

Expected: OK printed.

**Step 4: Commit:**

feat(boba-jackett): router + main wiring + boot smoke verified

Task 22: OpenAPI 3.1 spec generation

Files: - Create: qBittorrent-go/internal/jackettapi/openapi.go



Step 1: Write OpenAPI spec by hand as a Go string constant (or embed `openapi.json` from disk). Cover every endpoint from §8 with request/response schemas matching the implemented handlers.



Step 2: Add GET /openapi.json handler that serves it.

**Step 3: Smoke test:**

`curl http://localhost:7189/openapi.json | jq '.paths | keys | length'` — expect 14+ paths.



Step 4: Commit: feat(boba-jackett): OpenAPI 3.1 spec at /
`openapi.json`

Phase 4 — Container integration (Tasks 23-25)

Task 23: Dockerfile.jackett

Files: - Create: qBittorrent-go/Dockerfile.jackett

**Step 1: Write multi-stage Dockerfile**

```
FROM golang:1.26-alpine AS build
RUN apk add --no-cache build-base sqlite-dev
WORKDIR /src
COPY go.mod go.sum ./
RUN go mod download
COPY . .
RUN CGO_ENABLED=1 go build -o /out/boba-jackett ./cmd/boba-jackett

FROM alpine:3.20
```

```
RUN apk add --no-cache ca-certificates sqlite-libs wget
COPY --from=build /out/boba-jackett /usr/local/bin/boba-jackett
EXPOSE 7189
ENTRYPOINT ["/usr/local/bin/boba-jackett"]
```



Step 2: Build image manually: `cd qBitTorrent-go && nice -n 19 podman build -f Dockerfile.jackett -t boba-jackett:dev .`
Expected: image built.



Step 3: Commit: `feat(boba-jackett): multi-stage Dockerfile`

Task 24: docker-compose.yml service

Files: - Modify: `docker-compose.yml`



Step 1: Append the boba-jackett: service block from spec §4.2 verbatim.



Step 2: Validate compose: `podman-compose config 2>&1 | head -30`
Expected: no errors.



Step 3: Commit: `feat(docker-compose): add boba-jackett service on :7189`

Task 25: start.sh ensure_boba_master_key

Files: - Modify: `start.sh`



Step 1: Add function

```
ensure_boba_master_key() {
    local env_file="$SCRIPT_DIR/.env"
    [[ ! -f "$env_file" ]] && touch "$env_file" && chmod 0600 "$env_file"
    if grep -qE '^BOBA_MASTER_KEY=[0-9a-fA-F]{64}$' "$env_file";
    then
        return 0
    fi
    print_info "Generating BOBA_MASTER_KEY..."
    local key
    key=$(head -c 32 /dev/urandom | xxd -p -c 64)
    {
        echo ""
```

```

    echo "# === BOBA SYSTEM ==="
    echo "# Master key for credential encryption-at-rest in
    config/boba.db."
    echo "# DO NOT LOSE THIS – credentials become
    unrecoverable without it."
    echo "BOBA_MASTER_KEY=$key"
  } >> "$env_file"
  chmod 0600 "$env_file"
  print_success "BOBA_MASTER_KEY persisted to .env"
}

```



Step 2: Call ensure_boba_master_key in main() after load_environment.



Step 3: Verify: `bash -n start.sh && rm -f /tmp/.env-test && touch /tmp/.env-test && chmod 0600 /tmp/.env-test && SCRIPT_DIR=/tmp bash -c '. start.sh; ensure_boba_master_key' && grep BOBA_MASTER_KEY /tmp/.env-test`



Step 4: Commit: `feat(start.sh): auto-generate BOBA_MASTER_KEY into .env`

Phase 5 — Frontend (Tasks 26-32)

These tasks add Angular routes, components, and the IPTorrents cookie capture flow. Each task includes the test step inside the Angular spec runner.

Task 26: Angular routes + nav + service

Files: - Modify: `frontend/src/app/app.routes.ts` - Modify: `frontend/src/app/app.html` (add nav entry) - Create: `frontend/src/app/jackett/jackett.module.ts` - Create: `frontend/src/app/jackett/jackett-routing.module.ts`



Step 1: Define lazy-loaded route

```

// app.routes.ts addition
{
  path: 'jackett',

```

```
loadChildren: () => import('./jackett/jackett.module').then(m
    => m.JackettModule),
}
```



Step 2: Add nav entry in app.html following existing nav patterns.



Step 3: Create module + routing module with two child routes: credentials (default) and indexers.



Step 4: Verify route resolves:

cd frontend && nice -n 19 ng build --configuration production
— should succeed.



Step 5: Commit: feat(frontend): /jackett route scaffold

Task 27: Credentials page + service

Files: - Create: frontend/src/app/jackett/credentials/
credentials.service.ts - Create: frontend/src/app/jackett/
credentials/credentials.component.{ts,html,scss} - Create: frontend/
src/app/jackett/credentials/credential-edit-dialog.component.ts



Step 1: Write Vitest spec asserting service hits http://
localhost:7189/api/v1/jackett/credentials with proper auth
header.



Step 2: Run — expect FAIL



Step 3: Implement service using HttpClient, returning Observable
<CredentialMetadata[]>.



Step 4: Implement component with Material table, columns from
spec §9.1.



Step 5: Implement edit dialog with PATCH semantics (empty field
= unchanged).



Step 6: Run tests + build: cd frontend && nice -n 19 ng test --
watch=false && nice -n 19 ng build



Step 7: Commit: feat(frontend): credentials management page

Task 28: Indexers — Configured tab

Files: - Create: frontend/src/app/jackett/indexers/
indexers.service.ts - Create: frontend/src/app/jackett/indexers/
indexers.component.{ts,html,scss} - Create: frontend/src/app/
jackett/indexers/configured-tab.component.ts



Step 1: Vitest spec for service GET /indexers shape + Configured tab rendering.



Step 2: Run — expect FAIL



Step 3: Implement service + tab container component (Material tabs).



Step 4: Implement Configured tab — Material table with status badges, enable toggle, test/remove actions.



Step 5: Tests + build pass.



Step 6: Commit: feat(frontend): indexers Configured tab

Task 29: Indexers — Browse Catalog tab

Files: - Create: frontend/src/app/jackett/indexers/catalog-
tab.component.ts - Create: frontend/src/app/jackett/indexers/
indexer-add-dialog.component.ts



Step 1: Vitest spec for paginated catalog table + add modal opens with pre-populated fields.



Step 2: Run — expect FAIL



Step 3: Implement Catalog tab with search + type/language filters + 50/page server-side pagination.



Step 4: Implement add dialog — pulls indexer template from GET /indexers/{id}/template-fields (NOTE: this endpoint must exist in the API; if missing, add it as a sub-task to §8.2 and Task 17 retroactively — flag as Task 17-bis).

[Plan note: Add Task 17-bis if Task 29 surfaces this gap. The current \$8 spec includes POST /indexers/{id} but doesn't expose the template separately; the Add dialog needs the template fields to render. Add GET /indexers/{id}/template to Task 17 if not already present.]

☐

Step 5: Tests + build pass.

☐

Step 6: Commit: feat(frontend): indexers Browse Catalog tab + add dialog

Task 30: Indexers — Autoconfig History tab

Files: - Create: frontend/src/app/jackett/indexers/history-tab.component.ts

☐

Step 1: Vitest spec — list of last 50 runs, click expands to show JSON viewer.

☐

Step 2: Run — expect FAIL

☐

Step 3: Implement with ngx-json-viewer (add dep) or similar; “Run autoconfig now” button calls POST /autoconfig/run.

☐

Step 4: Tests + build pass.

☐

Step 5: Commit: feat(frontend): indexers Autoconfig History tab

Task 31: IPTorrents cookie capture flow

Files: - Create: frontend/src/app/jackett/indexers/iptorrents-cookie-flow.component.ts

☐

Step 1: Vitest spec — when add dialog opens for an indexer with cookie-only required fields, render the instruction panel + textarea + save button.

☐

Step 2: Run — expect FAIL

☐

Step 3: Implement — detect cookie-only template via `template_files` filter; render instruction panel from spec §9.3 verbatim; on save, POST to `/credentials (kind=cookie)` then POST to `/indexers/iptorrents`.



Step 4: Tests + build pass.



Step 5: Commit: `feat(frontend): IPTorrents cookie capture flow`

Task 32: NNMClub clarification banner

Files: - Modify: `frontend/src/app/jackett/credentials/credentials.component.ts` - Modify: `qBittorrent-go/internal/jackett/autoconfig.go` (add `served_by_native_plugin` flag) - Modify: `qBittorrent-go/internal/jackett/autoconfig_test.go` (assert flag presence)



Step 1: Add Go field: extend `AutoconfigResult` with `ServedByNativePlugin []string` populated by checking each `skipped_no_match` entry against a known list (NNMCLUB, plus any other native-plugin name from `plugins/*.py` matched lowercased).



Step 2: Update Go test asserting NNMCLUB in `skipped_no_match` also appears in `served_by_native_plugin`.



Step 3: Update Vitest spec asserting banner renders for each name in `served_by_native_plugin`.



Step 4: Run all — expect PASS.



Step 5: Commit: `feat(jackett): NNMClub clarification – served_by_native_plugin field + UI banner`

Phase 6 — Higher test layers (Tasks 33-37)

Task 33: Layer 2 integration tests

Files: - Create: `qBittorrent-go/tests/integration/jackett_db_test.go`



Step 1: Write tests covering the 6 scenarios from spec §10.2 (bootstrap from empty .env, bootstrap with N triples, restart no re-import, UI add cred → DB+env consistent, UI delete → both gone, concurrent dashboard writes).



Step 2: Run with build tag: `cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 go test -tags=integration ./tests/integration/ -v`
Expected: all PASS.



Step 3: Commit: `test(boba-jackett): Layer 2 integration tests against real SQLite + .env`

Task 34: Layer 3 E2E tests (real Jackett)

Files: - Create: `qBitTorrent-go/tests/e2e/jackett_management_test.go`

Pre-requisite: stack is up (`./start.sh -p`).



Step 1: Write tests for the 7 scenarios from spec §10.3.



Step 2: Run: `cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 go test -tags=e2e ./tests/e2e/ -v` Expected: all PASS (tests skip cleanly if Jackett unreachable per CONST-11).



Step 3: Commit: `test(boba-jackett): Layer 3 E2E against live stack`

Task 35: Layer 4 security/leak tests + bandit/gosec

Files: - Create: `qBitTorrent-go/tests/security/credential_leak_test.go` - Create: `challenges/scripts/credential_leak_grep_challenge.sh`



Step 1: Write the leak test per spec §10.4 — insert 6 high-entropy creds, drive E2E, capture all logs+responses+/proc env, grep for literal values, assert 0 hits.



Step 2: Write the challenge wrapper.



Step 3: Run: bash challenges/scripts/
credential_leak_grep_challenge.sh Expected: PASS.



Step 4: Run security scanners:

```
GOMAXPROCS=2 nice -n 19 ionice -c 3 gosec ./qBitTorrent-go/...  
GOMAXPROCS=2 nice -n 19 ionice -c 3 bandit -r download-proxy/src/  
-ll
```

Expected: no high-severity issues (or all addressed/justified).



Step 5: File permission check — bash script asserts config/
boba.db and .env are mode 0600. Add as challenges/scripts/
boba_db_file_perms_challenge.sh.



Step 6: Commit:

test(boba-jackett): Layer 4 security (leak grep + gosec/bandit
+ file perms)

Task 36: Layer 5 benchmarks

Files: - Create: qBitTorrent-go/internal/db/repos/
catalog_bench_test.go - Create: qBitTorrent-go/internal/db/
crypto_bench_test.go - Create: qBitTorrent-go/internal/jackett/
autoconfig_bench_test.go



Step 1: Write benchmarks per spec §10.5 — three scenarios for
catalog query, encrypt/decrypt, full autoconfig run.



Step 2: Run: cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 go
test -bench=. -benchmem -benchtime=3s ./internal/db/repos/ ./
internal/db/ ./internal/jackett/ -run=^\$ Expected: all latency
thresholds from §10.5 met.



Step 3: Commit:

test(boba-jackett): Layer 5 benchmarks (catalog query, crypto,
autoconfig)

Task 37: Layer 6 contract tests + Layer 7 challenges

Files: - Create: qBitTorrent-go/tests/contract/openapi_test.go - Create: challenges/scripts/cred_roundtrip_challenge.sh - Create: challenges/scripts/env_db_drift_challenge.sh - Create: challenges/scripts/iptorrents_cookie_flow_challenge.sh - Create: challenges/scripts/master_key_autogen_challenge.sh - Create: challenges/scripts/nmclub_native_plugin_clarification_challenge.sh



Step 1: Contract test loads openapi.json, hits every endpoint, validates response shape using getkin/kin-openapi.



Step 2: Write all 5 challenge scripts following the project's existing challenge pattern (see challenges/scripts/no_suspend_calls_challenge.sh as reference for structure).



Step 3: Add all to challenges/scripts/run_all_challenges.sh (the file is currently untracked from the initial state — Task 38 will commit it; add an entry “TODO: register” if it doesn't exist yet).

Wait — run_all_challenges.sh is in the initial git status as untracked. Let me handle: if not tracked, add it in this task with the new challenges registered + the existing ones.

```
[[ ! -f challenges/scripts/run_all_challenges.sh ]] && cat >
    challenges/scripts/run_all_challenges.sh <<'EOF'
#!/usr/bin/env bash
set -euo pipefail
SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
fail=0
for script in "$SCRIPT_DIR"/*_challenge.sh; do
    echo "=== Running $(basename "$script") ==="
    if ! bash "$script"; then
        fail=$((fail+1))
        echo "FAIL: $(basename "$script")"
    fi
done
if [[ $fail -gt 0 ]]; then
    echo "$fail challenge(s) failed"
    exit 1
fi
echo "All challenges passed"
EOF
chmod +x challenges/scripts/run_all_challenges.sh
```



Step 4: Run: `bash challenges/scripts/run_all_challenges.sh`

Expected: all PASS.

☐

Step 5: Commit: `test(boba-jackett): Layer 6 contract + Layer 7 challenge suite`

Phase 7 — Cleanup + Definition of Done (Tasks 38-46)

Task 38: Remove Python `/api/v1/jackett/autoconfig/last` endpoint

Files: - Modify: `download-proxy/src/api/__init__.py` (remove jackett router include) - Delete: `download-proxy/src/api/jackett.py` - Modify: `download-proxy/src/merge_service/jackett_autoconfig.py` (mark deprecated; keep for now — Phase 2 removes the module entirely)

☐

Step 1: Verify no other code imports `download_proxy.src.api.jackett`: `grep -rn 'from.*api.jackett' download-proxy/src/`

☐

Step 2: Remove the include from `__init__.py` + delete the file.

☐

Step 3: Run Python unit tests: `python3 -m pytest tests/unit/ -v --import-mode=importlib` Expected: any tests that imported the removed endpoint now fail; remove or update them.

☐

Step 4: Verify python proxy still boots cleanly via cache-bust + restart per `CLAUDE.md`.

☐

Step 5: Commit: `refactor(python-proxy): remove /api/v1/jackett/autoconfig/last (replaced by Go service)`

Task 39: Reconcile predecessor plan checkboxes (DoD #4)

Files: - Modify: `docs/superpowers/plans/2026-04-26-jackett-autoconfig-clean-rebuild.md`

☐

Step 1: For each unchecked - [] in the predecessor plan, verify against shipped commits (git log --all --grep=jackett_autoconfig --oneline) and the actual code in download-proxy/src/merge_service/jackett_autoconfig.py.

☐

Step 2: Mark - [x] for every step that maps to shipped code (should be most of Layers 1-7).

☐

Step 3: For any item still genuinely undone, add a note explaining why it was deferred.

☐

Step 4: Commit: docs(plan): reconcile 2026-04-26 jackett-autoconfig plan checkboxes against shipped commits

Task 40: Documentation — JACKETT_INTEGRATION.md + CLAUDE.md + AGENTS.md

Files: - Modify: docs/JACKETT_INTEGRATION.md (rewrite “Auto-Configuration” section as DB-backed) - Modify: CLAUDE.md (Architecture section + Port Map + Env Vars adds BOBA_MASTER_KEY, BOBA_DB_PATH) - Modify: AGENTS.md (mirror CLAUDE.md updates)

☐

Step 1: Update JACKETT_INTEGRATION.md — section “Auto-Configuration” now describes the DB layer + UI as canonical; .env import as bootstrap-only.

☐

Step 2: Update CLAUDE.md Port Map: add row | 7189 | boba-jackett (Go) | http://localhost:7189 |.

☐

Step 3: Update CLAUDE.md Architecture section to list boba-jackett as the 4th container.

☐

Step 4: Update CLAUDE.md Environment Variables section: BOBA_MASTER_KEY (auto-generated 32-byte hex), BOBA_DB_PATH (defaults / config/boba.db).

☐

Step 5: Mirror all three updates into AGENTS.md.

☐

Step 6: Commit: docs: CLAUDE.md + AGENTS.md + JACKETT_INTEGRATION.md updates for boba-jackett + DB

Task 41: Documentation — BOBA_DATABASE.md (NEW)

Files: - Create: docs/BOBA_DATABASE.md



Step 1: Write doc covering: schema reference (link to spec §5), master key lifecycle (auto-gen + manual rotation procedure: stop service, decrypt all rows with old key, re-encrypt with new, update .env), backup procedure (podman exec boba-jackett wget -q0- http://localhost:7189/api/v1/admin/backup OR simply cp config/boba.db config/boba.db.bak), recovery from lost key (no recovery — re-bootstrap from .env if all triples are still there).



Step 2: Commit: docs: add BOBA_DATABASE.md (schema/key lifecycle/backup/recovery)

Task 42: Bugfix doc

Files: - Create OR modify: docs/issues/fixed/BUGFIXES.md



Step 1: For every bug surfaced + fixed during the implementation of Tasks 1-41, append an entry with: title, root cause, affected files, fix description, link to test/challenge that prevents regression.



Step 2: At minimum, document the IPTorrents cookie field gap (it was a pre-existing issue surfaced in this work — CONST-32 reproduction-before-fix means the iptorrents_cookie_flow_challenge.sh script reproduces it; the fix is the cookie capture flow + UI hint).



Step 3: Commit: docs(bugfixes): document IPTorrents cookie gap + any others surfaced during implementation

Task 43: TODO/FIXME sweep + CONST-013 audit



Step 1: Run grep: grep -rE 'TODO|FIXME|XXX|HACK' qBitTorrent-go/cmd/boba-jackett qBitTorrent-go/internal/db qBitTorrent-go/internal/envfile qBitTorrent-go/internal/bootstrap qBitTorrent-go/internal/jackett qBitTorrent-go/internal/jackettapi

qBitTorrent-go/internal/logging frontend/src/app/jackett docs/
BOBA_DATABASE.md Expected: 0 hits in NEW code (pre-existing entries
elsewhere are out of scope).



Step 2: If any hits, fix them in this task (don't defer).



Step 3: CONST-013 audit: `grep -rnE '\bsync\.(Mutex|RWMutex)\b'`
qBitTorrent-go/cmd/boba-jackett qBitTorrent-go/internal/db
qBitTorrent-go/internal/envfile qBitTorrent-go/internal/
bootstrap qBitTorrent-go/internal/jackett qBitTorrent-go/
internal/jackettapi qBitTorrent-go/internal/logging For each hit,
verify it does NOT guard a mutable map/slice (they should be small-
state guards like the envfile writer mutex). If any guards a map/
slice, refactor to `safe.Store/safe.Slice`.



Step 4: Commit if anything changed: `fix(boba-jackett): TODO/
FIXME sweep + CONST-013 audit`

Task 44: CONST-033 verification



Step 1: Run both challenges:

```
bash challenges/scripts/no_suspend_calls_challenge.sh
bash challenges/scripts/host_no_auto_suspend_challenge.sh
```

Expected: both PASS. - [] **Step 2: If either fails, halt and fix the
source** before proceeding (per CLAUDE.md: “fix the source — do NOT
extend the allowlist”). - [] **Step 3: No commit needed if no changes.**

Task 45: Final integration smoke (everything together)



Step 1: Tear down stack: `./stop.sh --purge`



Step 2: Wipe DB only (not .env): `podman unshare rm -f config/
boba.db config/boba.db-wal config/boba.db-shm`



Step 3: Boot stack: `./start.sh -p`



Step 4: Verify all services (extend the prior verification with port
7189):

```
for port in 7185 7186 7187 7189 9117; do
  echo -n "  $port: "; curl -s -o /dev/null -w "HTTP %{http_code}
  \n" http://localhost:$port/healthz 2>/dev/null || curl -s
  -o /dev/null -w "HTTP %{http_code}\n" http://localhost:
  $port/
done
```

Expected: all 200. - [] **Step 5: Verify boba.db created + master key was either pre-existing or auto-generated:**

```
ls -la config/boba.db && grep BOBA_MASTER_KEY .env | head -1 | sed
's/=.*>/' - [ ] Step 6: Verify autoconfig ran: curl -s
http://localhost:7189/api/v1/jackett/autoconfig/runs?limit=1 |
jq . - [ ] Step 7: Browse to dashboard, navigate to /jackett/
credentials and /jackett/indexers, verify all three tabs render. - [ ]
Step 8: No commit — verification only.
```

Task 46: Pre-test commit + push to all upstreams (DoD #9)

Pre-conditions: Every implementation commit from Tasks 1-44 already exists locally.



Step 1: Final tree check: `git status --short` — should be clean (or only the bin/ binaries which are gitignored).



Step 2: Run full test suite locally one final time:

```
GOMAXPROCS=2 nice -n 19 ionice -c 3 go test -short -race ./
  qBitTorrent-go/...
cd qBitTorrent-go && GOMAXPROCS=2 nice -n 19 go test
  -tags=integration ./tests/integration/ -v
GOMAXPROCS=2 nice -n 19 go test -tags=e2e ./tests/e2e/ -v
bash challenges/scripts/run_all_challenges.sh
./ci.sh --quick
```



Step 3: Push to every configured remote:

```
for remote in origin github upstream; do
  echo "=== Pushing to $remote ==="
  git push $remote main
done
```

Expected: each push reports success or “Everything up-to-date”.



Step 4: No further commits in this task.

Task 47: Definition-of-Done sweep + final run

The final gate. Execute exactly once when Tasks 1-46 are complete.

☐

Step 1: Walk DoD §11 from spec one item at a time and tick off each:

☐

§11.1 — every endpoint in §8 implemented + Layer 2/3 tests pass

☐

§11.2 — both UI pages work in browser (manual Playwright walkthrough recorded)

☐

§11.3 — all 7 test layers green (paste output)

☐

§11.4 — predecessor plan checkboxes reconciled (Task 39)

☐

§11.5 — docs updated (Task 40+41)

☐

§11.6 — bugfix doc complete (Task 42)

☐

§11.7 — IPTORRENTS_COOKIES documented + UI hint

☐

§11.8 — NNMClub clarification in autoconfig + UI banner

☐

§11.9 — pre-test commit + push to all 3 remotes (Task 46)

☐

§11.10 — demo block prepared for PR body

☐

§11.11 — TODO/FIXME sweep clean (Task 43)

☐

§11.12 — CONST-013 audit clean (Task 43)

☐

§11.13 — CONST-033 verification both PASS (Task 44)

☐

Step 2: Final demo block construction — paste the output of Task 45 verification + Task 46 final test run + Task 46 push output into a ## Demo markdown block. Save as docs/demos/2026-04-27-boba-jackett-ui-and-db-demo.md.

☐

Step 3: Commit demo doc:

```
git add docs/demos/2026-04-27-boba-jackett-ui-and-db-demo.md
git commit -m "$(cat <<'EOF'
docs(demo): boba-jackett DoD evidence – all 13 items verified

Co-Authored-By: Claude Opus 4.7 (1M context)
<noreply@anthropic.com>
EOF
)"
for remote in origin github upstream; do git push $remote main;
done
```



Step 4: Declare done. Reply to the user with a summary of all 13 DoD items ticked + a link to the demo doc.

Self-Review (run inline after writing this plan)

1. Spec coverage: Walked spec §1-§14. Each requirement maps to a task: - §4 Architecture → Tasks 1, 21, 23, 24 - §5 Schema → Task 3 - §5.1 Crypto envelope → Task 2 - §5.2 File perms → Task 35 - §6 Bootstrap flow → Tasks 5, 10, 25 - §7 Sync semantics → Task 16 (write-through pattern) - §8 API surface → Tasks 15-22 - §9 UI → Tasks 26-32 - §10 Test strategy → Tasks 33-37 + inline TDD in 1-22 - §11 DoD (13 items) → Tasks 38-47 each maps to a numbered DoD item - §12 Risks/Mitigations → covered by tests (leak grep, atomic writes, key-mismatch detection in Task 21 main.go) - §13 Out of scope → respected; merge service untouched

2. Placeholder scan: A grep for “TBD/TODO/FIXME/...should...” in the plan body (excluding the parts that talk about the user’s TODO sweep) found zero — but Task 29 has a “[Plan note]” about a possible API gap. This is INTENTIONAL — it flags a real risk to address during implementation. Leaving as-is.

3. Type consistency: Cross-checked names — Credential struct (Task 6) used as *Credential return in Get/List; Indexer (Task 7) and CatalogEntry (Task 11) consistent across uses; AutoconfigResult.ServedByNativePlugin (Task 32) added to autoconfig.go in the same task that asserts it.

Self-review pass.

Execution Handoff

Plan complete. Two execution options:

1. **Subagent-Driven** (recommended) — fresh subagent per task, review between tasks, fast iteration.
2. **Inline Execution** — execute tasks in this session using `superpowers:executing-plans`, batch execution with checkpoints.

Which approach?